MIL-S-5733B(USAF)
30 June 1961
SUPERSEDING
MIL-S-5733A(USAF)
1 October 1952

MILITARY SPECIFICATION

SOLENOID, FUSE ARMING, TYPE F-1

1. SCOPE

1.1 This specification covers one type of fuse arming solenoid, designated as type F-1 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| QQ-P-416 | - | Plating, Cadmium (Electrodeposited). |
| QQ-Z-325 | - | Zinc Coating, Electrodeposited Requirements For. |
| PPP-B-566 | - | Boxes, Folding, Paperboard. |
| PPP-B-636 | - | Box, Fiberboard. |
| PPP-B-676 | - | Boxes, Setup. |
| PPP-T-60 | - | Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging. |
| PPP-T-76 | - | Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing). |

MILITARY

| | | |
|---|---|---|
| MIL-P-116 | - | Preservation, Methods of. |
| MIL-E-5272 | - | Environmental Testing, Aeronautical and Associated Equipment, General Specification For. |
| MIL-L-7870 | - | Lubricating Oil, General Purpose, Low Temperature. |
| MIL-C-45662 | - | Calibration System Requirements. |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-129 | - | Marking for Shipment and Storage. |
| MIL-STD-143 | - | Standards and Specifications, Order of Precedence for the Selection of. |
| MIL-STD-147 | - | Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base. |
| MIL-STD-1285 | - | Marking of Electrical and Electronic Parts. |

DRAWINGS

US AIR FORCE

| | | |
|---|---|---|
| 50C26144 | - | Solenoid Assembly - Fuse Arming, Type F-1. |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5945

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:34Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Service.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D.C. 20402.)

3. REQUIREMENTS

3.1 First article. First article solenoids shall be furnished for the inspections specified in 4.4.

3.2 Drawings. The solenoid requirements shall be as specified herein and in accordance with US Air Force Drawing 50C26144.

3.3 Materials. Specifications and standards for all materials, processes, parts, and Government certification and approval of processes and equipment which are not designated herein, and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143.

3.4 Design and construction. Solenoids shall be of the design, construction, weight, and physical dimensions specified. The complete type F-1 fuse arming solenoid shall consist of a cylindrical metal housing containing a coil assembly, movable armature, closure plates, support bracket, spring-loaded retainer, and an adjustable mounting clamp assembly. Solenoids shall be so designed to insure proper operation when mounted in any position. The solenoid shall be designed so that the adjustable spring-loaded retainer and the solenoid-operated locking pin for the arming wire shall be positioned apart and each shall function independently of the other. In operation, for an armed release, the application of current shall cause the locking pin to protrude across the opening of the retainer support thus retaining the arming wire. Deenergizing the solenoid then shall cause the locking pin to retract and thus permit the arming wire to fall free of the solenoid assembly.

3.4.1 Pressure adjustment. The manufacturer shall adjust the pressure adjustment screw to obtain the values referenced in figure 1. The adjustment screw shall be secured by a nut, and shall not require readjustment prior to completion of the tests specified herein.

3.4.2 Plating. Cadmium plating shall be in accordance with QQ-P-416, type I, class 2. Zinc plating shall be in accordance with QQ-Z-325.

3.4.3 Operation without lubricant. The mechanism shall be designed to operate without periodic lubrication. If an initial lubrication is necessary during assembly of the solenoid, it shall be limited to a slight wiping of the bearing surfaces with oil, in accordance with MIL-L-7870.

3.4.4 Threaded parts. All threaded parts shall be in accordance with Handbook H28. Where practical, all threads shall be in conformity with the coarse-thread series. The fine thread series may be used only for applications that show a definite advantage through their use. When a special diameter-pitch combination is required, the thread shall be of American National Form and of any pitch between 16 and 36, which is used in the fine-thread series.

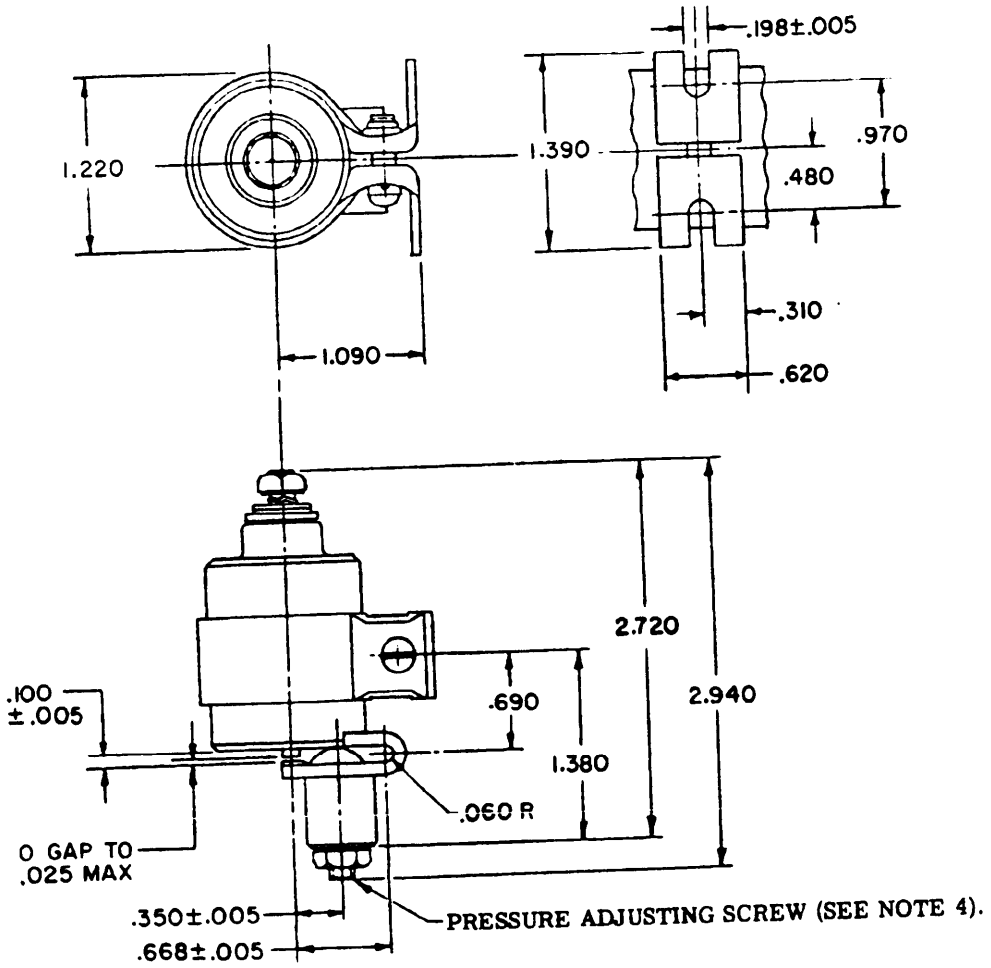
3.4.5 Dimensions. The overall dimensions shall be as shown on figure 1.

3.4.6 Mounting. The solenoid shall be mounted by means of an adjustable clamp bracket with two mounting holes as shown in figure 1.

3.4.7 Housing. The housing shall be dust-tight and moisture-proof.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:34Z
Check the source to verify that this is the current version before use.




| INCHES | MM | INCHES | MM |
|---|---|---|---|
| .025 | .64 | .690 | 17.53 |
| .060 | 1.52 | .970 | 24.64 |
| .100 | 2.54 | 1.090 | 27.69 |
| .198 | 5.03 | 1.220 | 30.99 |
| .310 | 7.87 | 1.380 | 35.05 |
| .350 | 8.89 | 1.390 | 35.31 |
| .480 | 12.19 | 2.720 | 69.09 |
| .620 | 15.75 | 2.940 | 74.68 |
| .668 | 16.97 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.010(.25 mm).
4. Screw shall be adjusted until the specified dimension is obtained.

FIGURE 1. Solenoid, fuse arming type F-1.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:34Z
Check the source to verify that this is the current version before use.

3.4.8 Coil. The coil shall be evenly wound with a kind and size of wire that will provide a solenoid conforming to the other requirements of this specification.

3.4.9 Weight. The weight of the solenoid shall not exceed 6.0 ounces.

3.5 Performance.

3.5.1 Voltage and current.

3.5.1.1 Operate voltage. When solenoids are tested as specified in 4.6.2.1, they shall operate without mechanical or electrical damage.

3.5.1.2 Reduced voltage. When solenoids are tested as specified in 4.6.2.2, they shall function properly.

3.5.1.3 Current. When solenoids are tested as specified in 4.6.2.3, the current drawn shall not exceed 0.35 ampere.

3.5.2 Load support (unlocked and locked).

3.5.2.1 Unlocked. When solenoids are tested as specified in 4.6.3.1, the spring-loaded retainer shall support a 5-pound weight without vibration, but shall not support a 7-pound weight.

3.5.2.2 Locked. When solenoids are tested as specified in 4.6.3.2, there shall be no evidence of mechanical damage.

3.5.3 Humidity. When tested as specified in 4.6.4, the operation of the solenoid shall not be impaired by exposure to an atmosphere of high relative humidity, and the voltage and current shall be as specified in 3.5.1.1 to 3.5.1.3 inclusive.

3.5.4 Altitude. When tested as specified in 4.6.5, solenoids shall withstand the specified pressure without damage and the voltage and current shall be as specified in 3.5.1.1 to 3.5.1.3 inclusive.

3.5.5 Vibration. When solenoids are tested as specified in 4.6.6, there shall be no evidence of mechanical damage.

3.5.6 Sand and dust. When tested as specified in 4.6.7, solenoids shall be capable of withstanding exposure to an atmosphere laden with sand and dust particles, and shall meet the following requirements:

| | |
|---|---|
| Operate voltage - - - - - - - - - | As specified in 3.5.1.1. |
| Reduced voltage - - - - - - - - - | As specified in 3.5.1.2. |
| Current - - - - - - - - - - - - - - | As specified in 3.5.1.3. |
| Load support - - - - - - - - - - - | As specified in 3.5.2. |

3.5.7 Salt spray. When tested as specified in 4.6.8, solenoids shall be capable of withstanding the salt laden atmosphere.

3.5.8 Fungus (mildew resistance). When tested as specified in 4.6.9, solenoids shall resist deterioration due to the action of fungi.

3.5.9 Torsion. When tested as specified in 4.6.10, there shall be no relative rotation of tested parts in excess of 0.020 inch, nor damage to the solenoid assembly.

3.5.10 Load (front closure plate). When tested as specified in 4.6.11, there shall be no looseness of the retainer in the support, nor any looseness of the front closure plate in the solenoid housing.

3.5.11 Torque (ground lead). When tested as specified in 4.6.12, there shall be no relative rotation of the base plate and solenoid housing, nor damage to the solenoid.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:34Z
Check the source to verify that this is the current version before use.

3.5.12 Armature pin protrusion. When solenoids are tested as specified in 4.6.13, the gap between the extended armature pin and the support bracket shall not exceed 0.025 inch (see figure 1).

3.5.13 Life. When tested as specified in 4.6.14, solenoids shall withstand 5,000 cycles of operation, and the voltage and current shall be as specified in 3.5.1.1 to 3.5.1.3 inclusive.

3.6 Marking. Solenoids shall be marked in accordance with method I of MIL-STD-1285 and shall include the following information, in the order shown:

Solenoid, fuse arming, type F-1.
Military part number M5733-1.
Manufacturer's part number.
Manufacturer's name, source code, or trademark.
Date code (inspection lot).

3.7 Workmanship. Solenoids shall be fabricated in such a manner as to be uniform in quality, and shall be free from cracked or displaced parts, sharp edges, burrs, and other defects that will affect life, serviceability, and appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.3).
(b) First article inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials used in fabricating the solenoids are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

4.4 First article inspection. First article inspection shall be performed by the supplier, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract or purchase order under which it is granted, unless extended by the Government to other contracts or purchase orders.

4.4.1 Sample size. Unless otherwise specified (see 6.2), one solenoid shall be subjected to first article inspection.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table I, in the order shown.



Source: https://assist.dla.mil -- Downloaded: 202610-17T05:34Z
Check the source to verify that this is the current version before use.

4.4.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

TABLE I. Inspection of solenoids.

| Examination or test | Requirement paragraph | Method paragraph |
|---|---|---|
| Group I | | |
| Visual and mechanical examination | 3.2, 3.4, 3.6, and 3.7 | 4.6.1 |
| Group II | | |
| Operate voltage | 3.5.1.1 | 4.6.2.1 |
| Reduced voltage | 3.5.1.2 | 4.6.2.2 |
| Current | 3.5.1.3 | 4.6.2.3 |
| Load support | 3.5.2 | 4.6.3 |
| Humidity | 3.5.3 | 4.6.4 |
| Altitude | 3.5.4 | 4.6.5 |
| Vibration | 3.5.5 | 4.6.6 |
| Sand and dust | 3.5.6 | 4.6.7 |
| Salt spray | 3.5.7 | 4.6.8 |
| Fungus (mildew resistance) | 3.5.8 | 4.6.9 |
| Torsion | 3.5.9 | 4.6.10 |
| Load (front closure plate) | 3.5.10 | 4.6.11 |
| Torque (ground lead) | 3.5.11 | 4.6.12 |
| Armature pin protrusion | 3.5.12 | 4.6.13 |
| Life | 3.5.13 | 4.6.14 |

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups I and II of table I.

4.5.2 Inspection lot. An inspection lot shall consist of all the solenoids utilizing the basic manufacturing process requirements, produced under essentially the same conditions, and offered for inspection at one time.

4.5.3 Sampling plan. Group I tests shall be performed on 100 percent of the solenoids furnished under this specification. Group II tests shall be performed on four solenoids from each inspection lot; no failures shall be allowed.

4.5.4 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit four additional sample units for inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.5 Noncompliance. If a sample fails to pass quality conformance inspection, the supplier shall take corrective action on the materials or processes, or both as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, quality conformance inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Final acceptance shall be withheld until the quality conformance inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failures and corrective action taken shall be furnished to the cognizant inspection activity.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:34Z
Check the source to verify that this is the current version before use.

4.5.6 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

4.6 Methods of examination and test.

4.6.1 Visual and mechanical examination (external). Solenoids shall be examined to verify that the materials, external design and construction, physical dimensions, weight, marking, and workmanship are in accordance with the applicable requirements (see 3.2, 3.4, 3.6, and 3.7).

4.6.2 Voltage and current.

4.6.2.1 Operate voltage (see 3.5.1.1). Solenoids shall be operated continuously for 2 hours on 24 to 28.5 volts dc, without damage.

4.6.2.2 Reduced voltage (see 3.5.1.2). Solenoids shall be connected in series with a resistance of 20 ohms and shall operate for 2 hours continuously without damage with 24 volts dc applied.

4.6.2.3 Current (see 3.5.1.3). Solenoids shall be energized with 24 volts dc and operated for 2 minutes continuously and shall not exceed the amperage rating specified in 3.5.1.3.

4.6.3 Load support (unlocked and locked) (see 3.5.2).

4.6.3.1 Unlocked (see 3.5.2.1). In the normal unlocked position, a 5- and 7-pound weight (on an arming wire swivel loop) shall be applied to the spring-loaded retainer. The load shall be imposed 10 times.

4.6.3.2 Locked (see 3.5.2.2). In the locked (solenoid energized) position, a maximum steady pull of 100 pounds at right angles to the axis of the solenoid plunger shall be applied with a standard arming wire loop. The locked solenoid plunger shall return to its unlocked position upon deenergization of the solenoid with a weight of 20 ounces applied to the plunger at a right angle to its axis using a standard arming wire loop. The load shall be imposed 10 times.

4.6.4 Humidity (see 3.5.3). Solenoids shall be subjected to the humidity test in accordance with procedure I of MIL-E-5272. Following this test, solenoids shall be subjected to operate voltage, reduced voltage, and current as specified in 4.6.2.1, 4.6.2.2, and 4.6.2.3, respectively.

4.6.5 Altitude (see 3.5.4). Solenoids shall be operated at 24 to 28.5 volts dc and shall be subjected to the altitude test in accordance with condition C, procedure VI of MIL-E-5272. Following this test, solenoids shall be subjected to operate voltage, reduced voltage, and current as specified in 4.6.2.1, 4.6.2.2, and 4.6.2.3, respectively.

4.6.6 Vibration (see 3.5.5). Solenoids shall be mounted on the vibration stand in the normal operating position and subjected to vibration for a period of 1 hour at any frequency between 17 to 50 Hz with an amplitude of .018 to .020 of an inch. During and after this test, the spring-loaded retainer shall support not less than a 2-1/4 pound weight, suspended by a standard arming wire loop. At the end of 1 hour, the weight shall be removed and the solenoid rotated 90 degrees from the previous position and vibrated for an additional hour at the frequency and amplitude specified above. Following this test, solenoids shall be visually examined for mechanical or electrical damage.

4.6.7 Sand and dust (see 3.5.6). Solenoids shall be subjected to the sand and dust test in accordance with procedure I of MIL-E-5272. Following this test, solenoids shall be subjected to operate voltage, reduced voltage, and current as specified in 4.6.2.1, 4.6.2.2, and 4.6.2.3, respectively. After several manipulations of the spring-loaded retainer, the solenoid shall then be subjected to the load support test specified in 4.6.3.

4.6.8 Salt spray (see 3.5.7). Solenoids shall be subjected to the salt spray test in accordance with procedure I of MIL-E-5272.



Source: https://assist.dla.mil -- Downloaded: 202610-17T05:34Z
Check the source to verify that this is the currentversion before use.

4.6.9 Fungus (mildew resistance) (see 3.5.8). Solenoids shall be subjected to the fungus resistance test in accordance with procedure I of MIL-E-5272.

4.6.10 Torsion (see 3.5.9). Solenoids shall be rigidly mounted so that the open end of the support plate assembly faces downward. Inscribe a line on the support and extend it onto the spring-loaded retainer housing. Apply a torque of 45 inch-pounds to the spring-loaded retainer housing.

4.6.11 Load (front closure plate (see 3.5.10). With solenoids mounted as specified in 4.6.10, insert a spacer of 0.100 inch thickness between the front closure plate and the spring-loaded retainer housing and gradually apply a load of 50 pounds vertically downward to the extreme outer end of the spring-loaded retainer housing.

4.6.12 Torque (ground lead) (see 3.5.11). The solenoid shall be rigidly mounted so that the thread terminal faces upward. Inscribe a line on the base plate and extend it across the rolled rim of the solenoid housing. Gradually apply a torque of 15 inch-pounds to the hub of the base plate.

4.6.13 Armature pin protrusion (see 3.5.12). Solenoids shall be fully energized and the gap between the extended armature pin and the supporting bracket shall be measured to determine conformance with limiting dimensions specified on figure 1.

4.6.14 Life (see 3.5.13). Solenoids shall be energized continuously at 28.5 volts dc, for a period of 10 hours without mechanical or electrical damage to the solenoids. Following this test, solenoids shall be subjected to operate voltage, reduced voltage, and current as specified in 4.6.2.1, 4.6.2.2, and 4.6.2.3, respectively.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Solenoids shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Solenoids shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Solenoids shall be individually packaged in accordance with MIL-P-116, submethod IC-3 insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. Each unit package shall be placed in a supplementary container conforming to PPP-B-566 or PPP-B-676.

5.1.1.5 Intermediate packaging. Not required.

5.1.2 Level C. Clean and dry solenoids shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. The packaged solenoids shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:34Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. The packaged solenoids shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged solenoids shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Solenoids, packed as specified in 5.2.1, shall be placed on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Solenoids, packed as specified in 5.2.2, shall be palletized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic.

5.2.4.3 Level C. Solenoids, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. Unit packages shall be marked in conformance with MIL-STD-1285. Also in addition to any special marking required by the contract or order, each unit package, supplementary and exterior container and unitized load shall be marked in accordance with MIL-STD-129. The nomenclature shall be solenoid; fuse arming, type F-1.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

6. NOTES

6.1 Intended use. The type F-1 fuse arming solenoid conforming to this specification is intended for use in bomber or fighter aircraft as a means of remotely controlling the arming of fused stores.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type number (see 1.1).
(c) List components by name and type designations as appearing on current parts control list.
(d) Number of first articles samples to be inspected and data required (see 4.4).
(e) Levels of preservation-packaging and packing (see 5.1 and 5.2).
(f) Special marking required (see 5.3).

Preparing activity:
Air Force - 80

Agent:
DSA - ES

(Project 5945-F216)



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:34Z
Check the source to verify that this is the current version before use.